

Section k: Implicit Differentiation

$$1. 2x + 2y \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}$$

$$2. 3x^2 - 3y^2 \frac{dy}{dx} = 6y + 6x \frac{dy}{dx}$$

$$\Rightarrow (-3y^2 - 6x) \frac{dy}{dx} = 6y - 3x^2$$

$$\Rightarrow (3y^2 + 6x) \frac{dy}{dx} = 3x^2 - 6y$$

$$\Rightarrow \frac{dy}{dx} = \frac{3x^2 - 6y}{3y^2 + 6x}$$

$$3. (2xy + x^2 \frac{dy}{dx}) + (3y^3 + 9xy^2 \frac{dy}{dx}) - 1 = 0$$

$$\Rightarrow (x^2 + 9xy^2) \frac{dy}{dx} = -3y^3 - 2xy + 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{-3y^3 - 2xy + 1}{x^2 + 9xy^2}$$

$$4. (3x^2y^2 + 2x^3y \frac{dy}{dx})$$

$$- (10xy + 5x^2 \frac{dy}{dx}) + 1 = 0$$

$$\Rightarrow (2x^3y - 5x^2) \frac{dy}{dx} = 10xy - 3x^2y^2 - 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{10xy - 3x^2y^2 - 1}{2x^3y - 5x^2}$$

$$5. 2x = \frac{(1 + \frac{dy}{dx})(x-y) - (x+y)(1 - \frac{dy}{dx})}{(x-y)^2}$$

$$\Rightarrow 2x(x-y)^2 = (x-y) + (x-y) \frac{dy}{dx} - (x+y) + (x+y) \frac{dy}{dx}$$

$$\Rightarrow 2x(x-y)^2 - (x-y) + (x+y) = [(x-y) + (x+y)] \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2x(x-y)^2 + 2y}{2x} = \frac{x(x-y)^2 + y}{x}$$

$$6. \frac{1}{2}(xy)^{-\frac{1}{2}}(y + x \frac{dy}{dx}) = \frac{dy}{dx}$$

$$\Rightarrow \frac{y + x \frac{dy}{dx}}{2\sqrt{xy}} = \frac{dy}{dx}$$

$$\Rightarrow y + x \frac{dy}{dx} = 2\sqrt{xy} \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{2\sqrt{xy} - x}$$

$$7. 35(x^2 + 3y^2)^{34}(2x + 6y \frac{dy}{dx}) = 1$$

$$\Rightarrow (35(x^2 + 3y^2)^{34} \cdot 6y) \frac{dy}{dx} = 1 - (35)(2x)(x^2 + 3y^2)^{34}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1 - (35)(2x)(x^2 + 3y^2)^{34}}{[35(x^2 + 3y^2)^{34}](6y)}$$

$$8. (-\sin(xy))(y + x \frac{dy}{dx}) = \frac{dy}{dx}$$

$$\Rightarrow (-x \sin(xy) - 1) \frac{dy}{dx} = y \sin(xy)$$

$$\Rightarrow (x \sin(xy) + 1) \frac{dy}{dx} = -y \sin(xy)$$

$$\Rightarrow \frac{dy}{dx} = \frac{-y \sin(xy)}{x \sin(xy) + 1}$$

$$9. \cos(x^2y^2)(2xy^2 + 2x^2y \frac{dy}{dx}) = 1$$

$$\Rightarrow [2x^2y \cos(x^2y^2)] \frac{dy}{dx} = 1 - 2xy^2 \cos(x^2y^2)$$

$$\Rightarrow \frac{dy}{dx} = \frac{1 - 2xy^2 \cos(x^2y^2)}{2x^2y \cos(x^2y^2)}$$

$$10. [3 \tan^2(xy^2 + y)] \sec^2(xy^2 + y)$$

$$(y^2 + 2xy \frac{dy}{dx} + \frac{dy}{dx}) = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{1 - 3y^2 \tan(xy^2 + y) \sec^2(xy^2 + y)}{(2xy + 1)(3 \tan^2(xy^2 + y) \sec^2(xy^2 + y))}$$

$$11. \frac{1}{2}(3 + \tan xy)^{-\frac{1}{2}} (\sec^2 xy) (y + x \frac{dy}{dx}) = 0$$

$$\Rightarrow \frac{x \sec^2 xy}{2\sqrt{3 + \tan xy}} \left(\frac{dy}{dx} \right) = \frac{-y \sec^2 xy}{2\sqrt{3 + \tan xy}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-y}{x}$$